CCCA Meta-Atlas — Bassez et al. 2021 (Breast)

Breast Cancer · 10x · 65,885 malignant cells (42 samples) → 65,464 analysed (41 samples, cell floor 17)

Do malignant cells recapitulate embryonic-development programs?

Non-branching (linear) manifold — single gastrulation/somitogenesis → neural-crest axis **GO / GO / NA**

GO / NO-GO GATES

Gate 1 — mixing & continuity **GO**

Harmony (θ=10, n_pcs=30) on sample, swept θ∈{2,5,10,20} × n_pcs∈{20,30,40} on a per-neighbourhood iLISI: median 4.9 → 8.6 of 41 samples (worst-neighbourhood p10 1.2 → 3.2), plateaus θ 2–10 then declines at θ=20. One connected component; 418 speck cells (0.63%) in 38 islands removed. Samples enter as fine intermixed speckle across one continuous, low-cycling body (3.1% cycling) — no per-patient archipelago → GO.

Gate 2 — module geography **GO (non-branching)**

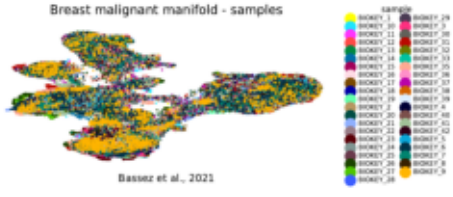
Single (Gast/pSom-origin) → Crest-terminus gradient. Smoothed geography: Gast~pSom −0.20, Crest~pSom −0.82, Crest~Gast −0.24. Along pseudotime pSom falls (ρ −0.27) and Crest rises (ρ +0.27); joint pSom&Crest enrichment 0.23 (<1 = inverse terminal gradient, not a co-expression hub). No Gast-enriched arm at any PPT stiffness. Present but not resolved into two competing arms → GO. Crest-arm inversion NOT flagged (Gast&Crest enrichment 0.50 <1; Crest tracks the terminus, not the Gast pole).

Gate 3 — cross-repression **NA**

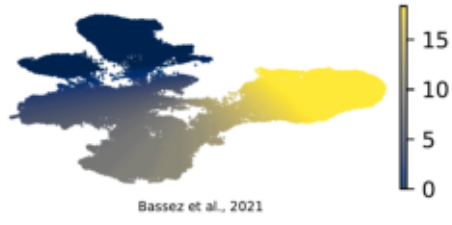
Non-branching topology (curve: 2 tips / 0 forks across N=15/20/30; PPT tree forks lack a Gast arm — max-tip Gast ≤ −0.001 vs Crest~pSom −0.87..−0.91, i.e. lobe over-fitting of one axis). No bifurcation to test → NA (branching-only 04_correlation / 05_modules notebooks not produced, per the deliverable spec).

STEP FIGURES

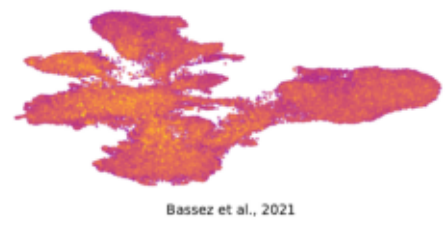
01 · Samples (Gate 1)



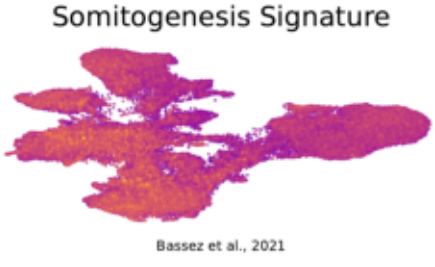
02 · Pseudotime



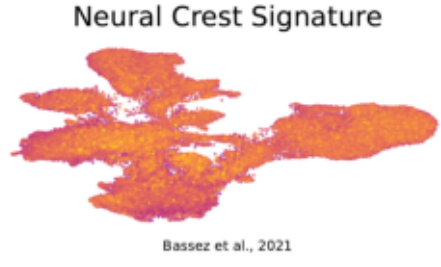
03 · Gastrulation



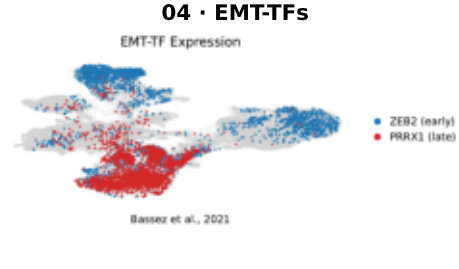
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



PARAMETER CHOICES & RATIONALE

Threshold	≥17 cells/sample = geometric mean across the dominant 92→3 spillover cliff (ratio 30.7, top/second 21.4). Dropped BIOKEY_34 (3 cells) → 41/42 samples, 65,882 cells (min 92/sample).
Embedding	PCA 50 comps, NO sc.pp.scale; Harmony θ=10 on sample; n_pcs=30, n_neighbors=30, UMAP spread=1.3; 2,498 overdispersed genes; gene recovery Gast 78 / pSom 93 / Crest 104 (union pool 231).
Trajectory	scFates curve (non-branching), Nodes=20, epg_λ=0.3, epg_μ=100; fold-free (self-approach 2.43, measured on the fitted graph; continuity corr(x_lin, t)=+0.93).
Root	tip 2 = Gast/pSom-high, Crest-low curve tip (developmental origin; Gast/pSom EARLY vs Crest LATE); axis not proliferation-driven (pseudotime vs cycling ρ +0.17).
Geometry	Linear-pSom/Crest — single axis; pSom globally strongest (raw means Gast −0.014, pSom +0.022, Crest −0.005).
EMT-TFs	ZEB2 (early/root) → PRRX1 (late/terminus), detection-gated from a 10-TF panel incl. FOXC/PRRX families (VIM excluded — a mesenchymal marker, not a TF).

Verdict The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching); no Gast↔pSom bifurcation, so Gate 3 is NA. Geometry Linear-pSom/Crest, consistent with the linear outcome across most CCCA datasets. G1 GO / G2 GO / G3 NA.